MIL-S-23821A(SHIPS)
20 September 1965
SUPERSEDING
MIL-S-23821(SHIPS)
2 September 1963
(See 6.4)

MILITARY SPECIFICATION

SIMULATOR OWN SHIP'S MOTION (OSMOS) (NAVAL SHIPBOARD)

1. SCOPE

1.1 This specification covers the design of an Own Ship's Motion Simulator (OSMOS) for use in Naval Ships for producing signals representing own ship's motion in various ships systems during training, casualty of gyro-compass or log, or when over-the-ground motion is known and should be provided.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-25 - Capacitors, Fixed, Paper-Dielectric, Direct Current, (Hermetically Sealed in Metallic Cases), General Specification for.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-I-15676 - Indicator's Ship's Course.
MIL-S-20708 - Synchros, 60 and 400 Cycle, General Specification.
MIL-S-22432 - Servo Motors, General Specification.

STANDARDS

MILITARY

MS17322 - Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle.
MIL-STD-701 - Preferred and Guidance List of Semiconductor Devices.

DRAWINGS

BUREAU OF SHIPS

9000-S6504-F-73687 - Standard Dial Markings for Interior Communication Order and Indicating Systems.

FSC 6930

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)
Standard 236.04 - Inspection of Fine-Pitch Gears (ASA B6.11-1956).

(Application for copies should be addressed to the American Gear Manufacturers Association, #1 Thomas Circle, N.W., Washington 5, D.C.)

AMERICAN STANDARDS ASSOCIATION (ASA)
B3.5-1960 - Tolerances for Ball and Roller Bearings.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 16, N.Y.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.3).

3.2 Description.- The OSMOS shall be a bulkhead mounted device, which in response to manually inserted rudder and speed orders shall realistically simulate the motion characteristics of the selected ship class and continuously transmit simulated own ship's course, own ship's speed and own ship's distance signals. The transmitted signals and front panel indicators shall be as specified in table I.

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Table I - Indicating and transmitting signals.

Signal	Indicate	Transmit
Rudder angle order	Dial (see 3.7.6.1)	None
Rudder angle position	Dial (See 3.7.6.2)	None
Own ship's course(OSC)	Dial (see 3.7.10)	31TX4 (10 degrees per rev.) 31TX4 (360 degrees per rev.) 31TX6 (10 degrees per rev.) 31TX6 (360 degrees per rev.)
Speed order	Dial (see 3.7.7.1)	None
Own ship's speed(OSS)	Counter (see 3.7.3.4)	31TX4 (10 knots per rev.) 31TX4 (40 knots per rev.) 31TX4 (100 knots per rev.) 31TX6 (10 knots per rev.) 31TX6 (40 knots per rev.) 31TX6 (100 knots per rev.)
Own ship's distance (OSD)	None	31TX6 (360 rev. per nautical mile) (240 r.p.m. at 40 knots)

3.2.1 Typical ship speed characteristics are as plotted on Figure 1. Ship's speed is plotted against time for the interval between the time and the speed order is executed and the time the ship reaches the ordered speed. The slope of the speed curve is representative of the ship's acceleration characteristics. The time to accelerate varies considerably for different classes of ships. For any particular ship class the time required to accelerate the ship to the ordered speed is a function of the ordered speed change.

3.2.2 All transmitters shall be 120 volts (v.), alternating current (a.c.), either 60 or 400 cycles (c.) as appropriate.

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3.3 Operation.- The operation of the OSMOS shall be as follows: Rudder angle and ship speed required shall be indicated when operator moves the manual controls. Suitable relays, or other devices, in the followup systems of the OSMOS shall duplicate the delays normally encountered in the ship when a rudder angle or ship speed order is given. Other devices shall simulate the acceleration and deceleration characteristics of a speed change and shall correct the turning rate of the ship for the specific input orders. In addition, the speed decay associated with various rudder angles shall be introduced into the speed output. The instantaneous values of the simulated OSC and OSS shall be continuously transmitted and displayed by the respective indicators. The rates of course and speed changes shall be determined by the tactical characteristics of the class of ship for which the equipment is adjusted. The devices which control the delays and rates of operation shall be adjustable to simulate varying tactical characteristics of different classes of ships.

3.4 General features.- The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein: (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern,)

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Dials and pointers
Dial sizes
Locking devices (see 3.5.3)
Washers
Ball bearings (precision grade 50 required) (see 3.5.5)
Parts - electrical - general
Requirements for semiconductor devices (see 3.5.18)

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Transients
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers
Relays
Synchros
Electrical tapes
Batteries
Dial illumination lamps
Switches
Indicator lights and lampholders
Fuses
Fuseholders and fuse clips
Metallic rectifiers
Printed wiring and circuits
Enclosures - general
Enclosure - accessibility (see 3.6.2)
Enclosure - degree of (dripproof) (see 3.6.1)
Enclosure - mounting
Stiffening grooves
Minimum sheet metal thicknesses (watertight enclosures)
Through bolting
Cable entrance
Ventilation
Size (for submarines)
Threaded devices
Rounded corners and edges
Internal subassembly protection (see 3.6.2)
Drilled and tapped holes
Welding
Temperature and humidity
Shock (see 3.5.22), vibration (nonvital vibration test required),
(inclination test is not required)
Primary power supply circuits
Power supply tolerances
Personnel protection
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards, connectors and terminals
Wiring
Dial illumination
Electrical insulation
Dielectric strength and insulation resistance clearances

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Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Bill of materials
Interchangeability and standardization (see 3.5.9)
Manuals
Repair parts
Repair parts (electronic)
Designation and marking (serial numbers are required)
Reports
Item names and nomenclature
Workmanship and general examination

3.5 Design.-

3.5.1 Excessive play between moving parts, resulting in noise when the equipment is subjected to vibration, shall be eliminated insofar as practicable.

3.5.2 The equipment and its operation shall not be adversely affected by magnetic fields up to 5 gausses.

3.5.3 External adjusting devices shall be free from excessive friction. Each adjusting device requiring a setting, shall be provided with a suitable graduate dial, scale, or indicator, which shall be so designed that it may be read without difficulty. Handles of adjusting devices shall be so constructed or protected that accidental contacts will not change their setting, or shall be provided with protective covers attached to the unit in such a manner as not to cause interference or affect ease of handling. All adjusting devices pertaining to ship's characteristics shall be provided with a locking device to prevent slippage between periodic resetting.

3.5.4 Resolvers, friction disks, lead screws, rollers and other mechanical components shall be of a suitable material, properly and evenly hardened or treated to prevent wear. They shall be provided with oil-saturated felt wipers for lubrication.

3.5.5 Bearings shall be double-shielded, shall be at least class ABEC3 of ASA Publication B3.5-1960, and shall be the prelubricated sealed-for-life type.

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3.5.6 Viewing windows, where required, shall be so mounted as to minimize danger of breakage or cracking due to stress at the edges.

3.5.7 Protective devices.- Protective devices such as mechanical stops, clutches, limit switches or similar devices shall be provided where necessary to prevent damage, operational malfunctions or the movement of ship's characteristic adjustments, if the OSMOS is permitted to exceed the maximum ranges of any parameters.

3.5.8 Parts of the equipment requiring accurate placing shall be marked or doweled, so that when disassembled for repairs or adjustments, the parts can be replaced in their proper position. Parts requiring occasional renewal due to unavoidable service wear, shall be designed for maximum practicable ease of replacement and alignment.

3.5.9 Interchangeability.- Except as specifically provided herein, similar parts and component assemblies, including repair parts or corresponding apparatus furnished on the same contract or order shall be interchangeable without the necessity of further machining or hand fitting of any kind.

3.5.10 Proposed changes in approved design affecting parts or assemblies, including repair parts, shall be specifically approved as satisfactory to the bureau or agency concerned before being made effective.

3.5.11 Terminal boards shall be provided with protective covers to prevent accidential short circuits while working on the equipment.

3.5.12 The effect of roll and pitch shall not adversely affect the operation of the equipment when tested in accordance with 4.4.2.

3.5.13 Provision shall be made to protect the OSMOS against possible damage in case the equipment is allowed to run to or beyond north or south latitude limits, or at speed inputs in excess of 40 knots. Mechanical limit stops shall be provided on the total speed and the component speed gear trains at both the zero-knot and the 40-knot positions to prevent exceeding the operating range of the OSMOS due to malfunction of speed input equipment.

3.5.14 Mercury and radium in any form, shall not be used in this equipment or in its manufacture or testing.

3.5.15 All cable shields shall be tied to a common point and grounded at that point within the equipment.

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3.5.16 Gears.- Gears used in critical backlash applications shall be at least sixty-four pitch and precision 1C as defined by AGMA Standard 236.04, and of a material satisfactory to the bureau or agency concerned. Corrosion resisting steel gears may be used where this material will provide superior performance. Low torque, high precision gears shall not be lubricated unless they are fully sealed in an enclosure.

3.5.17 Electron tubes. No electron tubes or thermionic devices of any type shall be used in this equipment.

3.5.18 Semiconductor devices.- Semiconductor devices shall be selected from MIL-STD-701. The contractor shall submit a tentative semiconductor complement report to the bureau or agency concerned for approval as soon as the equipment design requirements have been determined. In the event a change in the complement of semiconductor is found necessary, it shall be reported within 30 days. In order to assure sufficient quantity for maintenance of equipment in the Naval service, the contractor shall furnish the bureau or agency with a final semiconductor complement report after the equipment design has been completed and before production has begun. These reports shall be submitted on the same form DD 816 as used for electron tubes, and shall be combined into a consolidated form DD 816 for each equipment.

3.5.19 Accuracies.- The error of the OSMOS shall not exceed the values shown below under all test conditions specified in 4.4:

(a)	Course error	0.1 degree
(b)	Speed error	0.1 knot
(c)	Track error	5.0 percent
(d)	Rudder effect on speed error	5.0 percent
(e)	Rate of turn error	10.0 percent

3.5.20 Error calculations.- The error calculations shall be based on comparisons of the inputs of the OSMOS as indicated on its OSC dial and speed outputs indicated on bureau or agency concerned approved supplier furnished test equipment.

3.5.20.1 Track error.- Track error is the error in the track produced by the OSMOS and is expressed by the following formula:

$$E_t = 100. \frac{(D_a - D_s)}{(D_a)}$$

Where:

E_t = Track error in percent
D_a = Theoretical tactical diameter as defined by the ship's characteristic adjustments.
D_s = Tactical diameter produced by OSMOS.

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Tactical diameter is defined as the distance gained to the right or left of the original course where a turn of 180 degrees has been completed.

3.5.20.2 Rudder effect on speed error.- Rudder effect on speed error is the error in output speed of the OSMOS during turns and is expressed by the following formula:

$$E_s = 100 \frac{(S_a - S_s)}{(S_a)}$$

Where:

E_s = Rudder effect on speed error in percent.

S_a = Theoretical tactical diameter as defined by the ship's characteristic adjustments.

S_s = Speed indicated by the OSMOS.

3.5.20.3 Rate of turn error.- Rate of error is the error in the rate of the track produced by the OSMOS and is expressed by the following formula:

$$E_R = 100 \frac{(T_a - T_s)}{(T_a)}$$

Where:

E_R = Rate of turn error.

T_a = Time to turn 90 degrees or 180 degrees in actual ship.

T_s = Time to turn 90 degrees or 180 degrees indicating by OSMOS.

3.5.21 The supplier shall furnish accurate information on weight and vertical center-of-gravity (v.c.g.). This information shall be required as follows (see 6.2):

(a) By notation of calculated weight and location of v.c.g. on the outline drawing initially submitted for approval.

(b) On all revisions of drawings which result from changes in the calculations, under (a) above.

(c) By notation of the actually measured weight and corrected location of v.c.g., made prior to shipment, on the final revision of drawings sent to the Government.

3.5.22 The ship's characteristic adjustments shall be located inside the OSMOS cabinet and shall be easily accessible.

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3.6 Mechanical features.-

3.6.1 Enclosure.- All parts required for the proper operation of the OSMOS shall be mounted in an enclosing case (degree of enclosure dripproof). Provision shall be made for the protection of moving parts from dust. Mounting and external wiring of the equipment shall be accomplished without removal of components from the enclosure and without exposing the gear to dust. Internal subassemblies shall not be supported by the enclosure. The enclosure shall be designed for mounting on a bulkhead.

3.6.2 Accessibility.- All units and subassemblies of the equipment shall be easily accessible for repair and adjustment. Protection of the internal subassemblies shall be provided in accordance with MIL-I-983. Suitable supports shall be furnished where necessary to allow repair work without the units resting on electrical components, electrical connections or on components that may be disarranged or otherwise damaged. Knurled knob captive screws with a screwdrive slot shall be used on all covers that require removal or opening for operating adjustments.

3.6.3 Controls and indicators.- The following operating controls and indicators shall be provided:

(a) Rudder angle control knob, heading slew switch, and indicator (see 3.7.6).
(b) OSS control knob, speed slew switch, and indicator (see 3.7.7).
(c) OSC indicator (see 3.7.10).
(d) Illumination control (see 3.7.11).
(e) Power switch (see 3.7.12).
(f) Blown fuse indicators (see 3.7.5).
(g) Time totalizing meter (see 3.7.14).

3.6.4 Lettering and marking.- The size and characteristics of lettering and markings for all dials and indicators shall be in accordance with Drawing 9000-S6504-F73687.

3.6.5 Weight and size.- The weight and size of the OSMOS shall be kept to a minimum consistent with reliability, and shall not exceed the following: weight - 150 pounds, width - 21 inches, height - 29 inches, depth - 14 inches.

3.7 Electrical features.-

3.7.1 Servo amplifiers.- The servo amplifiers where required shall operate on 60 c. or 400 c. synchro signals, as appropriate, shall be transistorized, and shall consist of the necessary items arranged on a removable chassis. The excitation lines for the synchro transmitters shall be appropriately fused within the OSMOS. The total power taken from this source shall not exceed 150 watts.

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The amplifier chassis shall be grounded to the case, and a B minus bus isolation from the chassis shall be provided for the servo amplifiers, which shall be positively connected (bonded) to the case of the OSMOS at one physical point only. Matching transformers shall be used between amplifier output and servo motor control field if the servo motor is not directly matched to the amplifier during manufacture. The servo motor shall not be modified to match the impedance of the amplifier. All servo trimming circuit shall be external to the amplifier package. Servo amplifiers shall have adjustable gain controls capable of adjusting the gain over a plus or minus 6 decibel (db) range.

3.7.2 Synchro unit.- The synchro unit shall consist of the input synchro control transformers, a two-phase servo motor, gear trains, and the output synchro signal generators to provide for the receipt of the input synchro signal from the master gyrocompass, and log, and the retransmission of the computed and converted synchro signals to control type synchro signal receivers as required. The plate supporting the synchro control transformers, a servo motor, and synchro signal generators shall be so arranged as to be removable in order to adjust the synchro unit for electrical zero. Unless otherwise specified, the synchro generators and control transformers shall be contractor furnished material in accordance with MIL-S-20708. The OSMOS shall be protected against damage or permanent misalignment due to an incorrectly phased or excessive input signals. Such protection shall take the form of positive servo stops or other means suitable to the design configurations.

3.7.2.1 Synchro capacitors.- The correct synchro capacitors shall be provided for all synchro control transformers. These capacitors shall be mounted as close to the control transformers as possible. The capacitor section shall be rated for 600 v. or greater and shall be in accordance with MIL-C-25. Each synchro capacitor shall have the capacitance of its three capacitor sections balanced within 1 percent, as measured by the divergence of the capacitance of any one leg from the arithmetical mean of the three.

3.7.3 Servo motors.- The servo motors shall be in accordance with the requirements of MIL-S-22432 including high impact shock and audible structureborne noise. Motors shall be equipped with sealed for life bearings.

3.7.4 Power supply.- The equipment shall operate from a power supply of three phase, 400 c., 115 v., a.c., Sixty c., 115 v., a.c., shall be accepted as an input to the equipment to energize the 60 c. synchro transmitters and for illumination.

3.7.5 Blown fuse indicators.- Blown fuse indicators shall be located on the left side of the OSMOS front panel to indicate loss of power supply and loss of excitation to each of the important circuits.

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3.7.6 Rudder angle system.- The rudder angle system shall consist of the rudder angle control and the rudder angle indicator. The rudder angle control shall transmit the input order to the rudder angle indicator via a rate adjustment which shall cause the indicator to move with a rate corresponding to that characteristic of the ship in which it is installed after a 5 plus or minus 2 second time delay. The rudder angle indicator shall continuously transmit rudder angle to the turn simulator and speed computer (see figure 4).

3.7.6.1 Rudder angle control knob.- The rudder angle control knob shall be located on the lower left side of the OSMOS front panel and on the same horizontal line as the OSS control knob. The rudder angle control knob shall be of a push to set design or equivalent to provide a starting point for the five second time delay (see 3.7.6) and shall be limited to a total of 160 degrees movement and shall simulate 40 degrees right rudder to 40 degrees left rudder. Clockwise rotation of the control knob shall produce right rudder (cause the ship's bow to move to the right). Zero rudder position of the control knob shall be vertical and up. The control knob shall be so designed so that it will not slip from its set position when released by the operator (see 3.5.3). Any change in rudder angle order in excess of one degree shall be considered as a change of rudder angle command.

3.7.6.1.1 Rudder angle control dial.- The rudder angle control dial shall be clearly graduated in one degree increments between zero and 20 degrees and in 5 degree increments between 20 and 40 degrees rudder angle. Each 5 degree marker shall be emphasized and clearly marked with the respective rudder angle. Dial graduations shall be on a radius of at least 2 inches.

3.7.6.2 Rudder angle indicator.- The rudder angle indicator shall be located above the rudder angle control knob and protected by the same glass window as the speed indicator. The rudder angle indicator shall be a dial type indicator and shall be clearly graduated in one degree increments from 40 degrees left rudder to 40 degrees right rudder. Each 5 degree marker shall be emphasized and clearly marked with the respective rudder angle. Dial graduations shall be on a radius of at least 2 inches. The rudder angle indicator shall have an adjustable setting rate from 2 degrees per second to 7 degrees per second. The simulated rudder angle as displayed on the rudder angle indicator shall be transmitted to the turn simulator and the speed computer.

3.7.6.3 Heading slew switch.- A heading slew switch shall be mounted immediately below the rudder angle control knob. The desired heading shall first be obtained on a digital readout indicator which shall be graduated in one tenth (0.1) degree increments. Once the desired setting is obtained on the indicator, the actual own ship's course setting is accomplished by actuating a push to set mechanism. The push to set mechanism may or may not be a part of the heading digital readout indicator. This switch shall override any signals from the rudder angle control knob and change the OSC indicator at a rate of 10 degrees per second if held in the actuated position.

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3.7.7 Own ship's speed system.- The OSS system shall consist of the speed control knob, the speed computer and the speed output indicator. The speed control knob shall transmit the input order signal to the speed computer whose output is shown on the speed output indicator. The speed computer shall provide three corrections to the speed control order. The first correction shall be a time delay of 5 plus or minus 2 seconds simulating the actual time lapse between the time the order to change speed is given and the time when the ship first experiences a change in velocity. The second correction shall occur after the time delay and shall simulate the acceleration-deceleration characteristics of the ship. The correction for rudder angle effect to the speed control order shall be introduced into the speed computer at a point beyond where both the time delay and change of velocity characteristics corrections are introduced since this correction is independent from these corrections but will be imposed simultaneously with them whenever a rudder angle is applied. After the above corrections have been made, the corrected speed control order shall be continuously transmitted to the OSS output indicator and the turn simulator (see figures 3 and 4).

3.7.7.1 Own ship's speed control knob.- The OSS control knob shall be located on the lower right side of the OSMOS front panel and on the same horizontal line as the rudder control knob. The OSS control knob shall be a push to set design or equivalent in order to provide a starting point for the five second time delay (see 3.7.7) and the total angular motion shall not exceed two hundred degrees, which shall correspond to a speed range of zero to forty knots. Rotation of the knob in a clockwise direction shall produce an increase in OSS. The knob shall be so designed that it will not slip from its set position when released by the operator.

3.7.7.2 Own ship's speed control dial.- The OSS control dial shall be clearly graduated in one knot increments and every five knot marker shall be emphasized and denoted with the respective speed (knots). Dial graduations shall be on a radius of at least two inches. In addition to the above dial markings the following markings shall be provided:

5 knot division	-	1/3 speed
10 knot division	-	2/3 speed
15 knot division	-	Standard speed
20 knot division	-	Full speed
25 knot division	-	Flank speed

The twenty knot position of the control knob shall be vertical. up and equidistant from the zero and forty knot position.

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3.7.7.3 Own ship's speed computer.- Any change in speed order in excess of one knot shall be considered as a change of speed command. The OSS computer, upon receipt of a change of speed order, shall first provide a 5 plus or minus 2 second time delay. After the five second delay, the simulated ship's speed shall begin to change in response to the speed order. The speed change shall follow a predetermined rate which simulates the actual characteristics of the ship's class (see figures 1 and 2). A rudder angle correction shall be applied after the acceleration-deceleration corrections. After all of the corrections to OSS have been made the corrected OSS is continuously transmitted to the OSS output indicator and turn simulator.

3.7.7.3.1 Acceleration-deceleration correction.- A change of own ship's speed is defined as any change in the speed command having a magnitude greater than one knot and less than forty knots and shall have the acceleration-deceleration characteristics shown on figures 1 and 2. The time required to accomplish the ordered speed change shall be adjustable within the limits set forth later. The ship's acceleration characteristic is represented by curve A-B, figure 1 and deceleration by curve C-D, figure 2. Points A and C represent the instants when the ship first experiences a velocity change (see 3.7.7). Points B and D represent the instants when the command speed is reached. Points A and C are independent of both the magnitude of speed change and the ship's velocity at the time of the speed change command. However points A and C shall be adjustable over one eighth of the time duration with respect to their positions on the curve so that the acceleration-deceleration characteristics may be provided which match the desired ship class characteristics. As the ship's speed changes to the command speed the rate of velocity change shall increase and then decrease in such a manner as to provide a smooth speed change having the same characteristics as shown on figures 1 and 2. All speed changes must follow the characteristic curves regardless of magnitude. The time required to make a forty knot speed change shall be adjustable within the limits of 60 to 800 seconds. The time required to make a one knot speed change shall be adjustable within the limits of 10 to 20 seconds. The time required for any speed change shall vary linearly between the one knot and 40 knot settings see figure 6. The ship's speed shall be continuously transmitted to the rudder angle correction portion of the ship's speed computer.

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3.7.7.3.2 Rudder angle correction.- The rudder angle correction of OSS shall be applied after the 5 second time delay and the acceleration-deceleration corrections have been made. All rudder angles other than zero degrees shall cause a decrease in OSS. The magnitude of this deceleration shall be adjustable and linear to both the instantaneous rudder angle and the instantaneous acceleration-deceleration corrected OSS. The maximum decrease in OSS shall not exceed 8 knots at 40 degrees rudder angle and 40 knots OSS (see figure 3). The acceleration and deceleration characteristics due to rudder angle shall be similar. The time required to accomplish this change in OSS shall be adjustable and linear to the amount of rudder angle. The response time adjustment shall be capable of covering a range of 10 seconds to 150 seconds for 40 degrees of rudder (see figure 3a).

3.7.7.4 Own ship's speed output indicator.- The OSS output indicator shall be located on the OSMOS front panel above the OSS control knot and on the same horizontal line as the rudder angle indicator. The same glass window shall protect the OSS output indicator and the rudder angle indicator. The indicating dial shall be a drum digital readout type, having a minimum 3/16 inch high characters, and shall be graduated in one tenth (0.1) knot increments over the range of zero to 40 knots. Each one tenth increments shall be clearly marked with its appropriate numeral. The setting rate of the OSS output indicator shall be greater than 3 knots per second. The OSS output indicator signals shall be transmitted in accordance with 3.2.

3.7.7.5 Own ship's speed manual input circuit.- The own ship's speed manual input circuit shall bypass the speed computer and provide a rapid means of setting the OSS output indicator. The OSS manual input controls shall be mounted on the lower right side of the OSMOS front panel and on the same horizontal line as the heading slew switch. The desired manual speed input shall first be obtained on a digital readout indicator which shall be graduated in one tenth (0.1) knot increments. Once the desired setting is obtained on the indicator, the actual OSS setting is accomplished by actuating a push to set mechanism. The push to set mechanism may or may not be a part of the OSS manual input indicator.

3.7.8 Own ship's distance output signal.- The OSS signal from the OSS output indicator shall be converted to an OSD signal and shall be continuously transmitted in accordance with 3.2.

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3.7.9 Turn simulator.- The turn simulator combines rudder angle, own ship's speed and the maneuvering characteristics of the ship to provide a continuous own ship's course signal to the OSC indicator. The maneuvering characteristic curve is the ship's turning curve, whose radius is measured perpendicular to the OSC and the radius magnitude is determined by the characteristic adjustment settings. OSS shall have no effect on the maneuvering characteristic curve, only establish how fast own ship moves along the curve thereby establishing OSC. The correct OSS as transmitted from the OSS computer is used for this function. Only a rudder angle change in excess of one degree shall constitute applied rudder. The maneuvering characteristic curve shall be defined by four adjustable points (1, 10, 25, 40 degrees RA). The maximum-minimum turning radius at the adjustable points are tabulated in table II and illustrated in figure 5. The magnitude of the ship's turning radius for a specific rudder angle between the given reference points is proportional to the turning radius of the adjacent reference points.

Table II - Turn simulator characteristics.

Rudder angle (degrees)	1	10	25	40
Maximum turning radius (yards)	3500	1200	900	500
Minimum turning radius (yards)	600	200	100	50

3.7.10 Own ship's course indicator.- The OSC indicator shall be located on the vertical centerline of the OSMOS front panel above the rudder angle indicator and the OSS indicator. The dial of the OSC indicator shall be the same as the type E or F ship's course indicators of MIL-I-15676 with the exception that the indicator shall function as a 1 and 36 speed indicator having an accuracy of 0.1 degree. The OSC indicator shall receive inputs from the turn simulator. The heading indicated by the OSC indicator shall be transmitted in accordance with 3.2.

3.7.11 Illumination control.- The illumination control shall be located in the lower left hand corner of the OSMOS front panel. All dials and counters shall be illuminated in red and shall be controlled by a single rheostat or other control device as specified in the paragraph of MIL-I-983 for dial illumination lamps.

3.7.12 Power switch.- A power switch shall be located in the lower left hand corner and below the illumination switch on the OSMOS front panel. The switch shall be a two-position switch (on-off) which breaks both sides of all power circuits in the equipment including illumination.

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3.7.13 Second command signal.- The OSMOS system shall be so designed that when a second speed or rudder angle command is initiated prior to completion of the first command, the first command shall continue toward completion during the second command five second time delay. Upon completion of the five second time delay the second order shall then override the first command using its instantaneous values as a starting point for the second command.

3.7.14 Time totalizing meter.- A time totalizing meter shall be mounted in clear view on the front panel to provide a means of telling when the OSMOS is to be calibrated. A meter similar to that described in MS17322 shall be used.

3.7.15 Effect of synchro bus load on synchro control transformers.- In order to prevent excessive reflected errors on the synchro bus, the secondary load on the synchro control transformers used in this equipment shall be greater than 15,000 ohms.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Additional inspection requirements.-

4.1.1.1 The supplier shall submit to the bureau or agency concerned for approval procedures to be followed during preproduction inspection (see 4.2) and quality conformation inspection (see 4.3).

4.1.1.2 The accuracy tests of 4.4.1 shall be performed after all of the tests and examinations listed in tables III, IV and V. In addition, the tests of 4.4.1 shall also be performed during the supply line voltage and frequency variation test, the temperature and humidity test, the roll and pitch test, the effects of magnetic fields test and the vibration test (see 4.1.1.2 and 4.4.1.6).

4.1.1.3 The general examination, shielding and radio frequency noise test, insulation resistance test and the shock test may be excepted from the requirements of 4.1.1.1.

4.1.1.4 The results of the accuracy tests required by 4.1.1.1 shall meet the requirements of 3.5.19.

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4.2 Preproduction inspection.- One sample OSMOS shall be subjected to the examination and tests specified in table III, which should be performed in the order listed.

Table III - Preproduction inspection.

Test	Requirement paragraph	Test paragraph
General examination	3.4	MIL-I-983
Accuracy	3.5.19	4.4.1
Supply line voltage and frequency variation	3.4	MIL-I-983
Shielding and radio frequency noise	3.4	MIL-I-983
Temperature and humidity	3.4	MIL-I-983
Roll and pitch	3.5.12	4.4.2
Effects of magnetic fields	3.5.2	4.4.3
Dielectric strength	3.4	MIL-I-983
Insulation resistance	3.4	MIL-I-983
Enclosure	3.4	MIL-I-983
Vibration	3.4	MIL-I-983 and 4.4.4
Shock	3.4	MIL-I-983 and 4.4.5

4.3 Quality conformance inspection.-

4.3.1 Group A tests.- Each equipment offered for delivery on the contract or order shall be subjected to the group A tests specified in table IV.

Table IV - Group A tests.

Test	Requirement paragraph	Test paragraph
Group A tests		
General examination	3.4	MIL-I-983
Accuracy	3.5.19	4.4.1
Dielectric strength	3.4	MIL-I-983
Insulation resistance	3.4	MIL-I-983
Supply line voltage and frequency variation	3.4	MIL-I-983

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4.3.2 Group B tests.- Group B tests, specified in table V, shall be conducted on one complete equipment when the basic design of the equipment or the material of a vital part has been changed.

Table V - Group B tests.

Test	Requirement paragraph	Test paragraph
Group B tests		
Shielding and radio frequency noise	3.4	MIL-I-983
Temperature and humidity	3.4	MIL-I-983
Roll and pitch	3.5.12	4.4.2
Effects of magnetic fields	3.5.2	4.4.3
Enclosure	3.4	MIL-I-983
Vibration	3.4	MIL-I-983 and 4.4.4
Shock	3.4	MIL-I-983 and 4.4.5

4.4 Test procedures.-

4.4.1 Accuracy.- Accuracy tests shall be made with the output signal of OSS at 40 knots per revolution (60 c.) connected to an underwater log speed indicator, and the output signals of OSC at one- and thirty-six-speed (400 c.) and OSS at 10 and 100 knots per revolution (400 c.) connected to the supplier furnished test equipment. The outputs from the supplier furnished testing equipment of components of OSD North-South and East-West in the form of step-by-step signals at 750 turns per nautical mile, 115 v. direct current shall be connected to the supplier furnished test equipment.

4.4.1.1 Course error test.- Quality conformance test for accuracy of course output shall consist of determing the amount of error resulting with a constant speed of fifteen knots and course inputs of each ten degrees of heading from 0 degree through 350 degrees. Readings shall be taken after the OSMOS is settled on course and there is no indication of change of course on the OSC dial of the supplier furnished test equipment course error of each reading shall not exceed the limits of 3.5.19(a).

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4.4.1.2 Speed error test.- Quality conformance test for accuracy of speed output shall consist of determining the amount of error resulting during eight runs of constant course. A run shall be made for an OSC of every 45 degrees from zero through 315 degrees of heading. During each run readings shall be taken at each of the following speeds in the order listed: 3, 10, 18, 25, 35, 27, 15 and 5 knots. Readings shall be taken after the OSMOS is settled on speed and there is no indication of change of speed on the underwater log speed indicator. Error in speed of each reading shall not exceed the limits of 3.5.19(b).

4.4.1.3 Track error test.- Track error shall be determined by applying a specific amount of rudder angle, or order, to the OSMOS from an initial condition of constant course and speed to produce a track on the supplier furnished test equipment of a tactical curve representing a complete 180 degrees of course change. The semicircular curves produced by the OSMOS on the supplier furnished test equipment shall be superimposed on a predetermined theoretical tactical curve, and the deviation from this curve is defined as the track error (see 3.5.20.1). The track error test shall consist of 16 runs listed in table VI. The track error of each run shall not exceed the limits of 3.5.19(c).

Table IV - Track error test.

Run	Initial course	Direction of turn	Initial speed	Amount of rudder angle
1	000	Right	12	15
2	000	Right	12	20
3	000	Right	12	25
4	000	Right	12	35
5	090	Left	17	15
6	090	Left	17	20
7	090	Left	17	25
8	090	Left	17	35
9	180	Right	22	15
10	180	Right	22	20
11	180	Right	22	25
12	180	Right	22	35
13	270	Left	27	15
14	270	Left	27	20
15	270	Left	27	25
16	270	Left	27	35

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4.4.1.4 Rudder effect on speed error test.- Rudder effect on speed shall be determined by observing the reduction of speed with constant rudder angle and an initial speed as listed in table VI. The readings shall be taken from the underwater log speed indicator readings when heading has changed 90 degrees and 180 degrees from the initial heading. The amount of the error shall be determined as specified in 3.5.20.2 using the tactical data provided. The rudder effect on the speed error test may be conducted concurrently with the track error test. The rudder effect on speed error of each run shall not exceed the limits of 3.5.19(d).

4.4.1.5 Rate of turn error test.- Rate of turn error test shall be determined by observing the time required for each change of heading of 90 degrees and 180 degrees during the turns required in the track error test. The time shall be measured from the instant of insertion of the course change into the OSMOS by the operator and until the course has changed 90 degrees and 180 degrees as indicated on the OSC dial of the DRAI. The amount of the error shall be determined as specified in 3.5.20.3 using the tactical data provided. The rate of turn error of each run shall not exceed the limits of 3.5.19(e).

4.4.1.6 When accuracy tests are required during the testing prescribed by 4.4.1 the following abbreviated accuracy test may be used (see 4.4.3 for exceptions):

(a) Course error test shall be run as described in 4.4.1.1 except that errors shall be determined for course inputs of each 40 degrees of heading from 0 degrees through 320 degrees.
(b) Speed error test shall be run as described in 4.4.1.2 except that during each run the following speeds may be omitted 10, 18, 27, and 5 knots.
(c) Runs 1, 3, 6, 7, 12, 14, and 15 of table VI may be omitted during the track error test, the rudder effect on speed error test, and the rate of turn error test.

4.4.2 Effect of roll and pitch.-

4.4.2.1 The purpose of this test shall be to determine whether the OSMOS will operate satisfactorily under conditions of roll and pitch.

4.4.2.2 The unit under test shall be mounted on a Scorsby test stand which shall be operated with an amplitude of roll of 40 degrees on each side of the vertical with a period between 8 and 10 seconds, and an amplitude of pitch of 10 degrees on each side of the vertical with a period between 6 and 8 seconds during which the abbreviated accuracy tests of 4.4.1.6 shall be conducted. The unit may be mounted in any one or more positions in which it may be required to operate aboard ship and the roll and pitch axes of the Scorsby test stand may be interchanged, if necessary, to provide clearance for the motion of the unit.

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4.4.3 Effects of magnetic fields.- The accuracy tests of 4.4.1 shall be made to ensure that a magnetic field will not affect the accuracy of the OSMOS as specified in 3.5.2. The abbreviated accuracy test of 4.4.1.6 shall not be used for this test.

4.4.4 Vibration.- The OSMOS shall conform to the vibration test for non-vital equipment requirements specified in MIL-I-983.

4.4.5 Shock.- The OSMOS shall conform to the high impact (HI) shock requirements of MIL-S-901 for class II, Grade B requirements.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Simulator, Own Ship's Motion.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Preparation for delivery requirements for onboard repair parts shall be as specified in the contract or order for the repair parts.

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5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 the following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging, packing and marking.- The equipment, accessories, repair parts and manuals shall be preserved and packaged level A or C; packed level A, B or C; and marked as specified in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 Bid data.- Bidders shall be required to furnish the following:

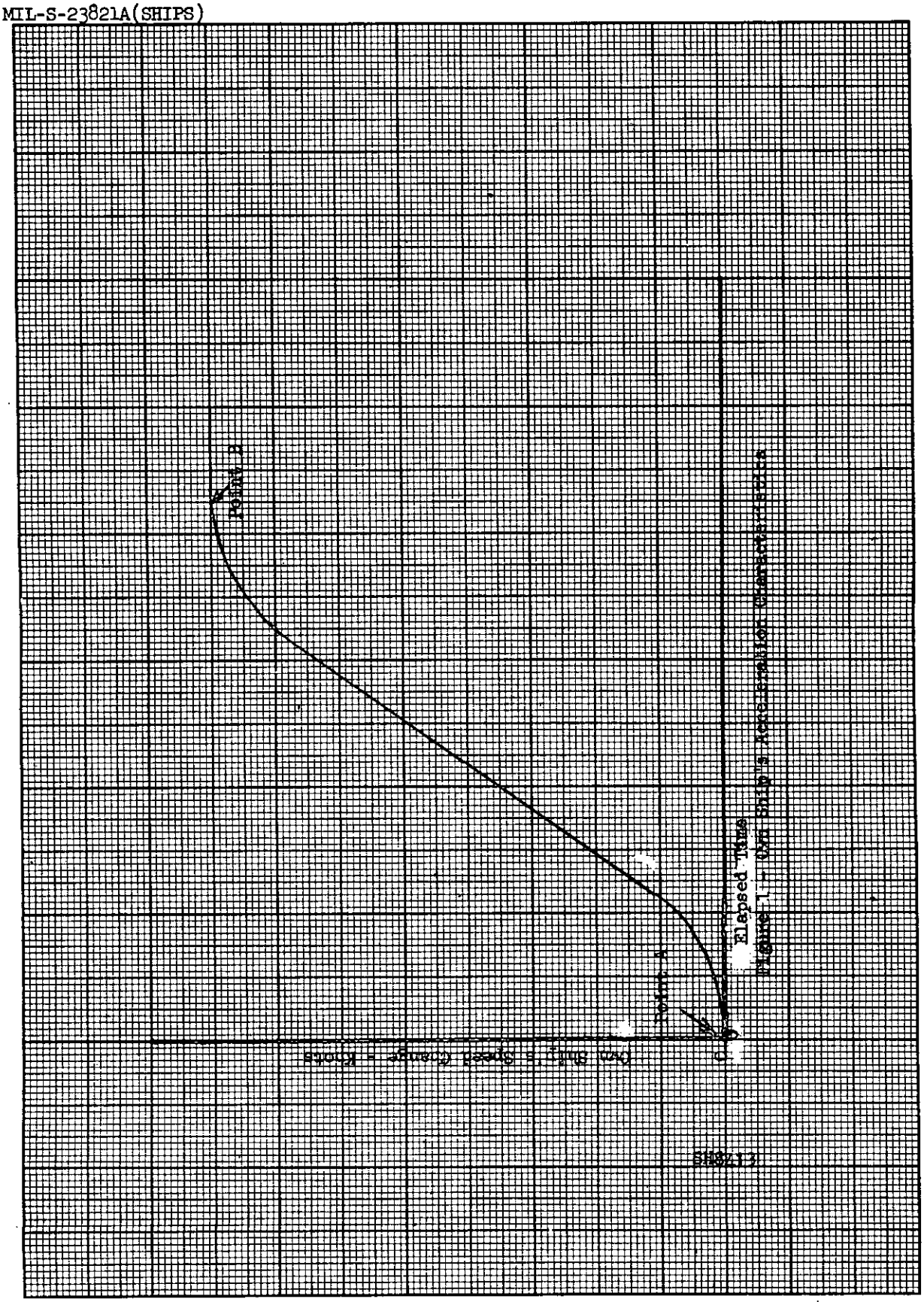
(a) Information on weight and v.c.g. of each unit of equipment weighing 100 pounds or more (see 3.5.21).
(b) Items known to be stowed or group located in quantity so as to constitute an aggregate weight of 100 pounds or more.

6.3 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

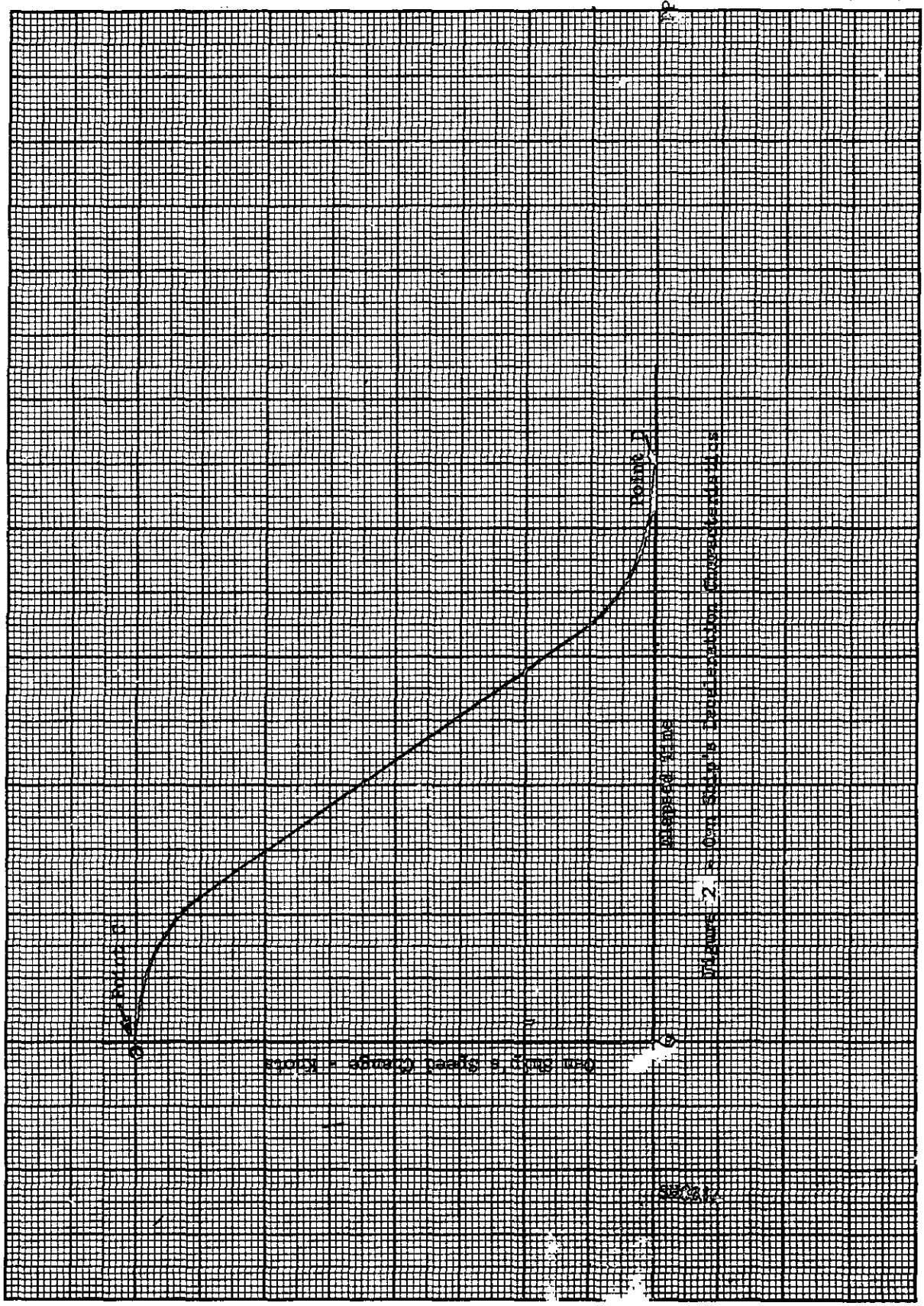
6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
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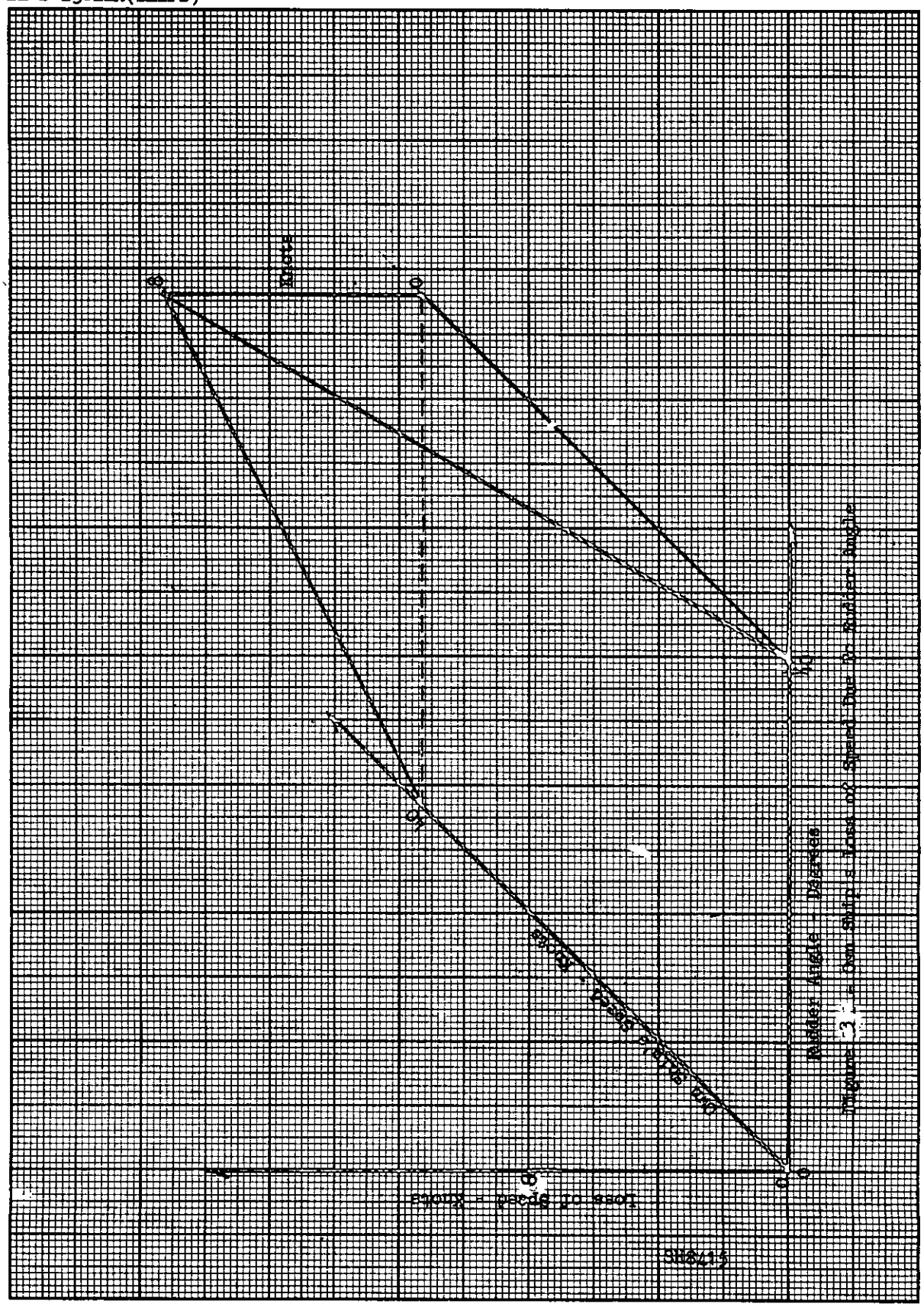


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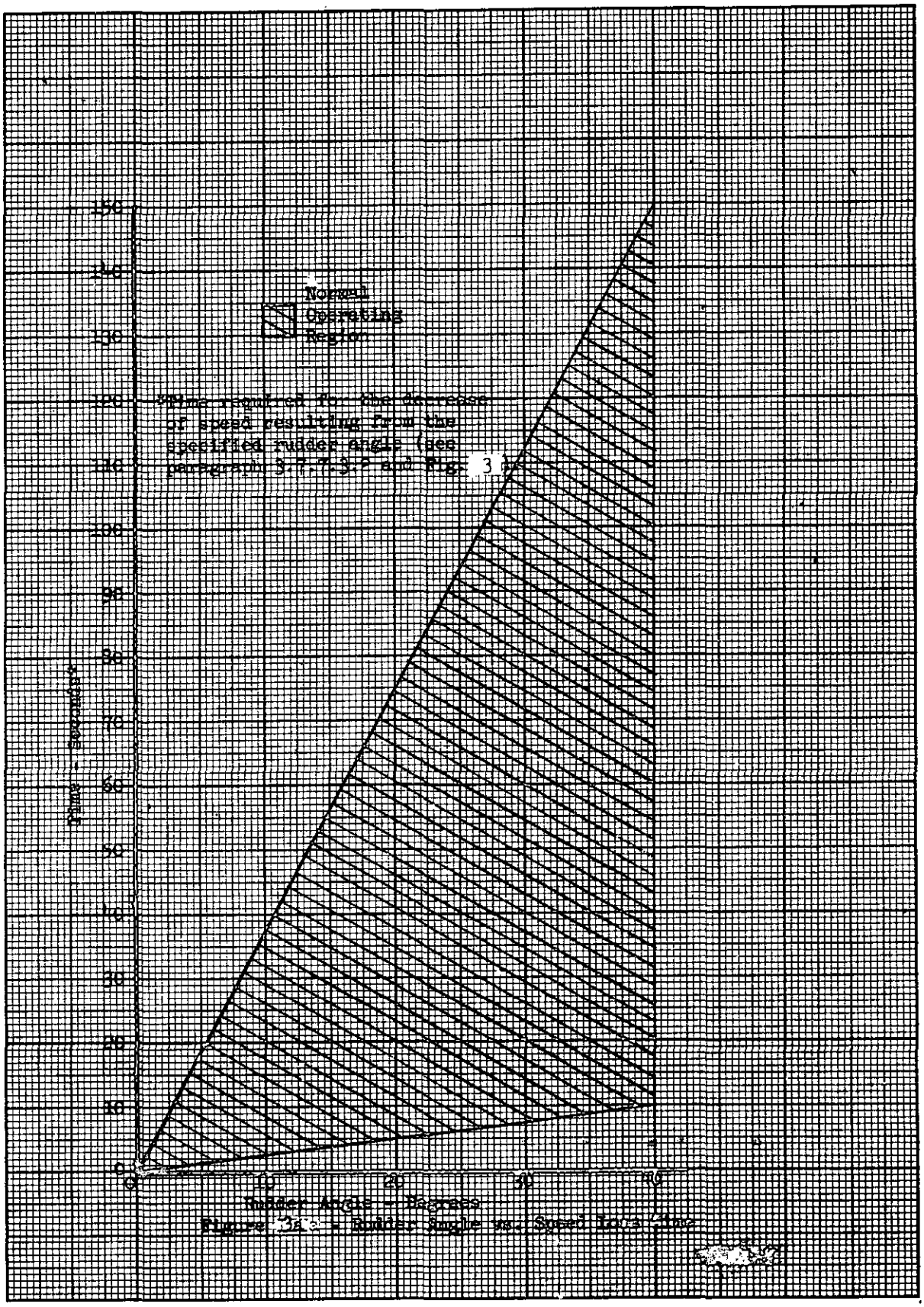


Figure 3a. - Rudder Angle vs. Speed Lever Time

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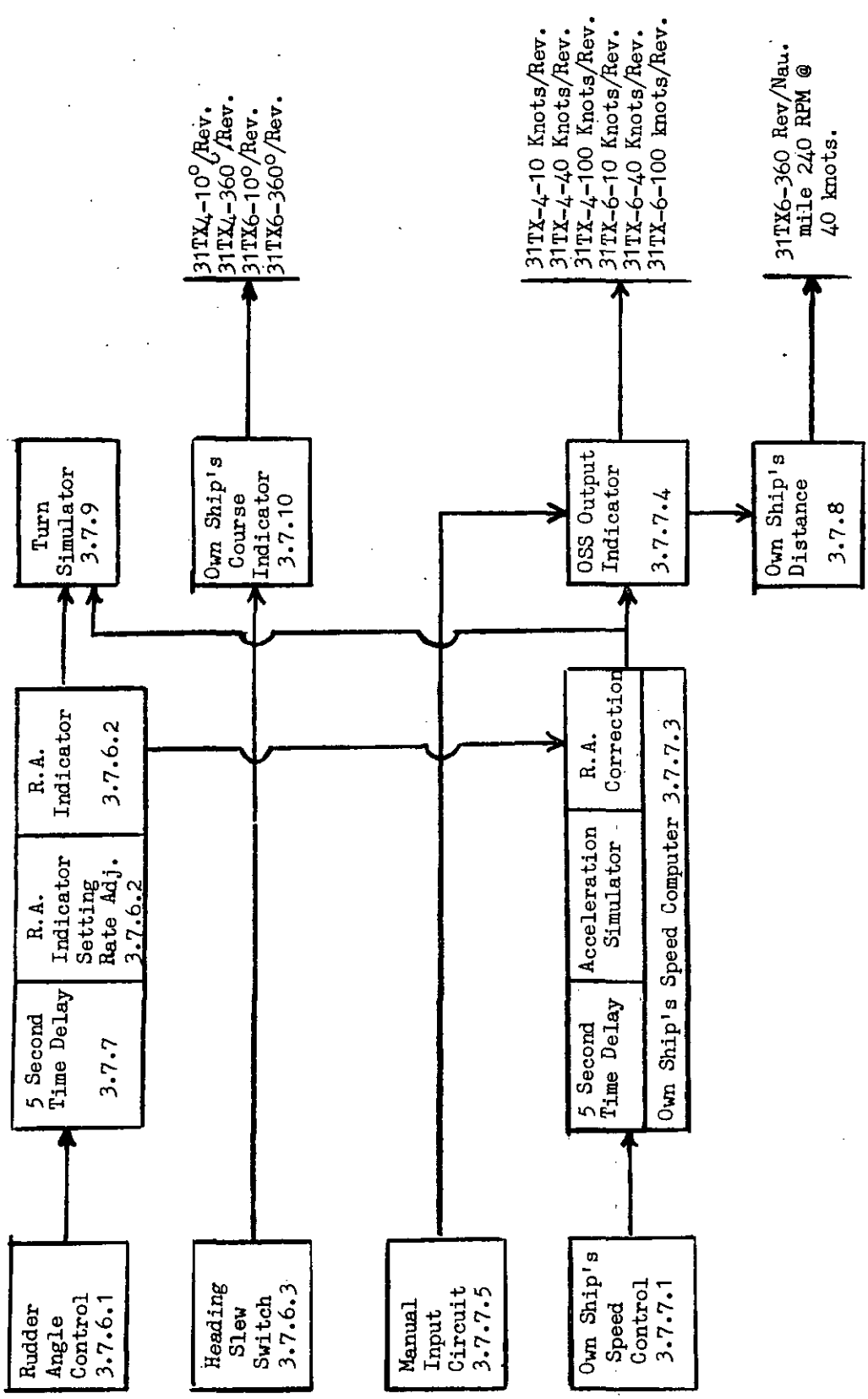
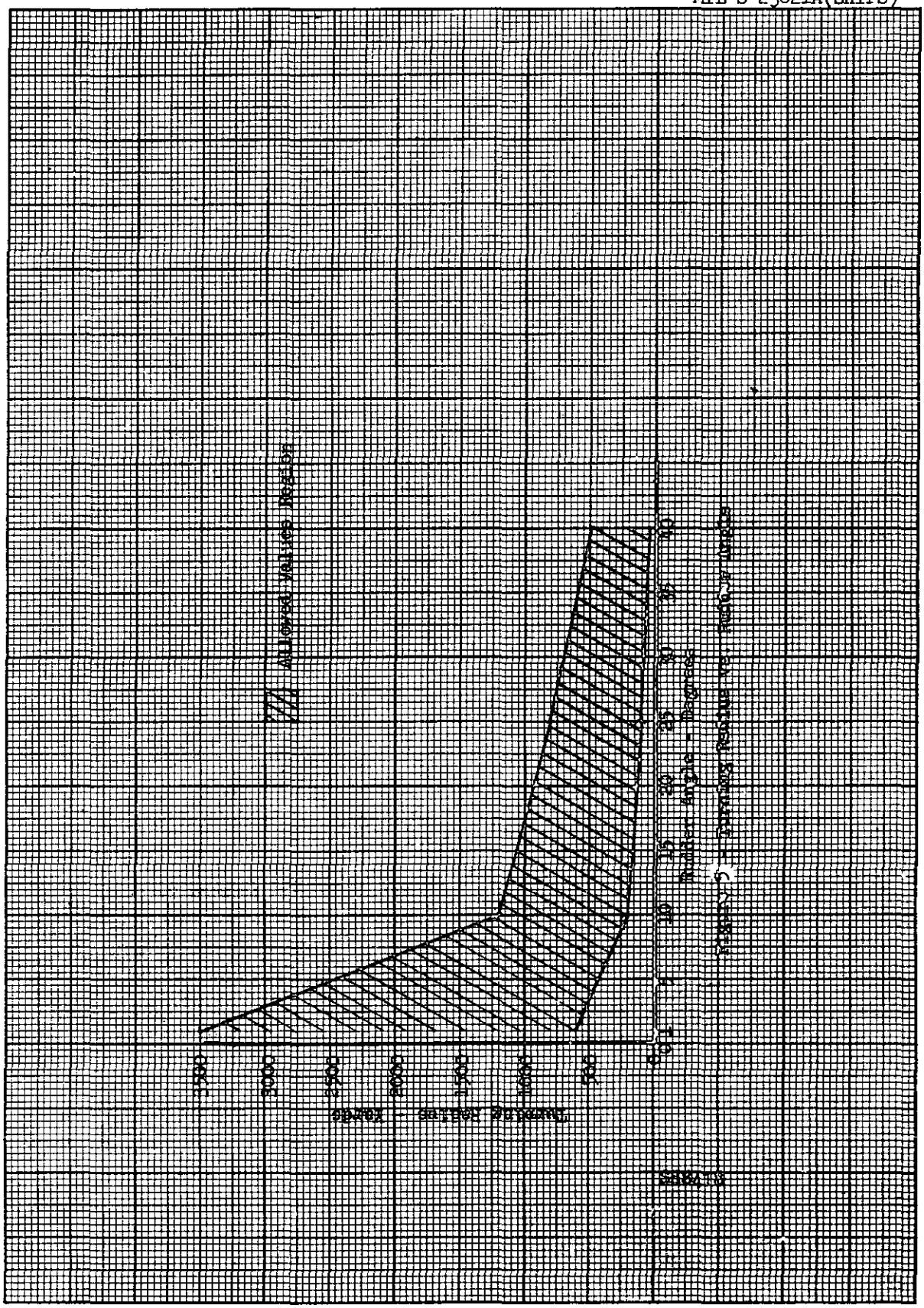


Figure 4 - Simulator Own Ship's Motion (OSMOS)(Naval Shipboard)

SH8417

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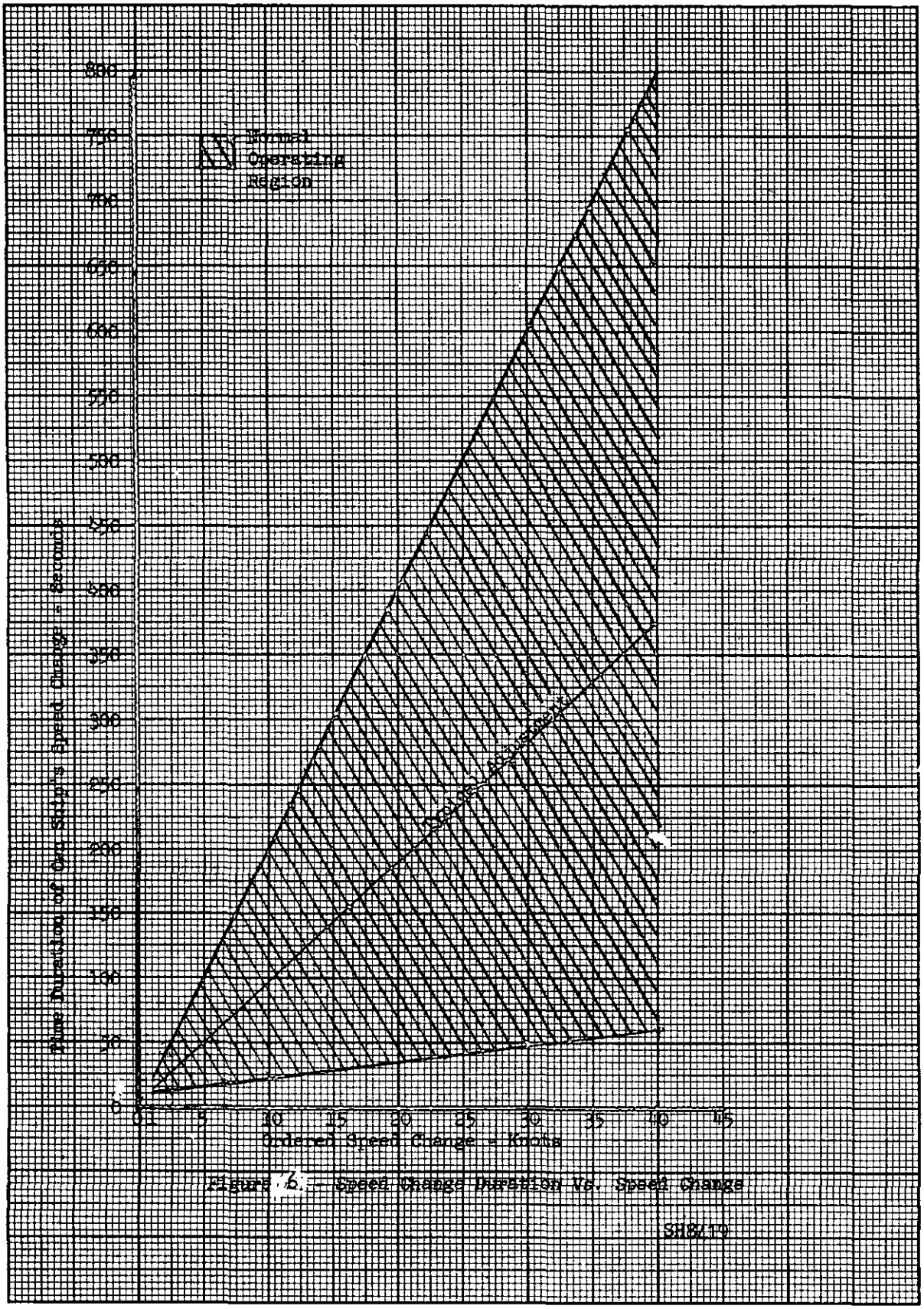


Figure 6. — Speed Change Duration Vs. Speed Change

SHEET 9

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BUREAU OF SHIPS
WASHINGTON, D. C. 20360

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DEPARTMENT OF THE NAVY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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